Hallam Land Management, Taylor Wimpey & Hannick Homes

Eastern Villages, Swindon

GREAT CRESTED NEWT MITIGATION STRATEGY

December 2015

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1.0 INTRODUCTION

1.1 This report has been prepared by FPCR Environment and Design Ltd on behalf of a consortium comprising Hallam Land Management Ltd, Taylor Wimpey and Hannick homes to provide details regarding the presence of great crested newts within and around a site to the east of Swindon. This report presents the results of a great crested newts (GCN) survey and mitigation strategy to prevent adverse impacts on this species.

Site Context

1.2 The Study Area is centred on the Ordnance Survey Grid Reference SU 2007 8752 and lies to the east of Swindon close to the village of South Marston.

1.3 The site is primarily used for agriculture at present, with pastoral fields bisected by hedgerow field boundaries in the west, to the south of South Marston, and a range of agricultural uses centred around Rowborough Farm to the east, including arable use, turf production and biomass fuel production.

1.4 The site and wider area to the south has been the focus of a wide range of surveys from at least 1999, including a number completed by FPCR. These surveys included amphibian survey over a number of years both within the site and wider Eastern Villages Allocated site area.

Legislation and Policy Guidance

1.5 GCN, their habitats in water and on land are protected under Section 41 of the Conservation of Habitats and Species Regulations 2010 and under the Wildlife and Countryside Act 1981 (as amended). Taken together, these make it an offence to:

a. deliberately capture or intentionally take a GCN;

b. deliberately or intentionally kill or injure a GCN;

c. to be in possession or control of any live or dead GCN or any part of, or anything derived from a GCN;

d. damage or destroy a breeding site or resting place of a GCN;

e. Intentionally or recklessly obstruct access to any place that a GCN uses for shelter or protection;

f. intentionally or recklessly disturb a GCN while it is occupying a structure or place that it uses for shelter or protection;

g. deliberately disturb any GCN, in particular any disturbance which is likely to

   i. impair their ability to survive, breed, reproduce or to rear or nurture their young; or in the case of hibernating or migratory species, to hibernate or migrate; or

   ii. to affect significantly the local distribution or abundance of the species to which they belong.

1.6 Although the law provides strict protection to GCN, it also allows derogation from this protection under Section 53 of the Conservation of Habitats and Species Regulations 2010 through the

issuing of EPS licences for development works. These licences in England are currently determined by Natural England (NE).

1.7 Where a lawful operation is required to be carried out, which is likely to result in one of the above offences, an EPS licence may be obtained from NE to allow the operation to proceed.

1.8 As part of the licence application process a number of 'Tests' have to be met by the application.

1.9 Natural England Guidance Note: European Protected Species and the Planning Process – Natural England's Application of the 'Three Tests' to Licence Applications (March 2011) states:

"In determining whether or not to grant a licence Natural England must apply the requirements of Regulation 53[5] of the Regulations and, in particular, the three tests set out in sub-paragraphs (2)(e), (9)(a) and (9)(b)[6]. (1) ***Regulation 53(2)(e****) states: a licence can be granted for the purposes of "preserving public health or public safety or other imperative reasons of overriding public interest including those of a social or economic nature and beneficial consequences of primary importance for the environment".*

(2) ***Regulation 53(9)(a)*** *states: the appropriate authority shall not grant a licence unless they are satisfied "that there is no satisfactory alternative".*

(3) ***Regulation 53(9)(b)*** *states: the appropriate authority shall not grant a licence unless they are satisfied "that the action authorised will not be detrimental to the maintenance of the population of the species concerned at a favourable conservation status in their natural range."*

1.10 Conservation status is defined as "the sum of the influences acting on the species concerned that may affect the long term distribution and abundance of its population within its territory". It is assessed as favourable when:

- population dynamics data on the species concerned indicate that it is maintaining itself on a long term basis as a viable component of its natural habitats, and
- The natural range of the species is neither being reduced nor is likely to be reduced for the foreseeable future, and
- There is, or will probably continue to be, a sufficiently large habitat to maintain its populations on a long term basis.

1.11 These tests must not only reach agreement with Natural England when assessing a Licence application they must also be assessed by the planning authority when determining a planning application.

2.0 SURVEY METHODOLOGY

Existing Information

2.1 Historically, the site and wider Eastern Villages allocated site has been subject to a number of amphibian surveys since 2009 and the presence of GCN known within the local area. This data, some of which was collected by FPCR has been reviewed in order to provide additional context.

2.2 In particular, survey data for ponds within the site have been reviewed. These surveys were conducted in 2002, 2006, 2009 & 2012.

Field Survey

Habitat Suitability Index (HSI)

2.3 Descriptions of accessible water bodies within 250 m of the site, where not isolated by running water, busy roads or other barriers to dispersal, were completed and assessed for potential to support an amphibian population, including GCNs, using the Habitat Suitability Index (HSI)[1]. This assesses ponds against ten pre-determined criteria, which produces a score which indicates the suitability for GCN occupation. Criteria include habitat types, vegetation cover, geographical location and proximity to other water bodies.

2.4 Ponds reviewed were mainly confined to the site boundaries, however a further two ponds located in the adjacent nightingale wood were also visited and found to be dry in 2012 and subsequently in June 2013.

Aquatic Surveys

2.5 Aquatic surveys were conducted 2015 at the six water bodies identified in Figure 1, following the methods recommended by Natural England and as detailed in the GCN Mitigation Guidelines (English Nature, 2001). Natural England, in their Standing Advice, suggests that in order to accurately inform the extent to which a protected species is affected by development surveys of medium to high impact schemes should not be more than 2 – 3 years out of date and as a result surveys are considered up-to-date. All surveys were completed during appropriate weather as identified in Table 1 below:

Table 1: Survey weather conditions

Survey Date	Survey Conditions						
	Turbidity	Wind	Rain	Evening temp		Morning temp	
				Air pm	Water pm	Air am	Water am
09/04/2015	1	1	0	19	10	13	10
20/04/2015	1	0	0	16	12	13	11
07/05/2015	1	0	0	15	10	10	10
14/05/2015	2	2	0	14	11	11	10
04/06/2015	1	1	0	15	12	11	11
08/06/2015	1	0	0	16	12	14	11

2.6 To determine the presence or absence of GCNs, four individual survey visits were performed in previous years a population size class assessment was made and comprised a total of six visits. However, due to the now extensive data available over a number of years, any further population size class assessment is not considered necessary. On each survey occasion, three of a possible four different techniques (egg search, sweep net, bottle-trap and torch) were used where possible. A summary of each is provided below:

[1] Refer to Oldham et al (2000)

Bottle Trapping

2.7 Bottle traps were set within the waterbody in the evening at densities of one trap per two metres of shoreline (where feasible) and left overnight for inspection in the morning. Traps were partially submerged in the water leaving an air bubble in the bottle and secured by a cane marked with a high visibility tape to ensure relocation the following day. Care was taken to ensure that trapping did not occur during excessively warm weather, when the temperature inside the trap could rise considerably, reducing oxygen levels and potentially suffocating the newts.

Sweep Netting

2.8 Long handled sweep-nets were used to sample the margins of the pond for GCNs, with approximately 15 minutes of netting per 50 m of shoreline.

Torching

2.9 Torching involved searching the waterbody after dusk using high-powered torches (Clulite CB-2) to scan the margins and potential display areas for newts. The perimeter of the pond was walked slowly recording any newts observed. Torch surveys are unsuitable within heavily vegetated and/or turbid ponds or after periods of heavy rain as visibility is diminished.

Egg Searching

2.10 Newts lay single eggs on leaves of aquatic plants or other suitable pliable material, after which the material is folded over the egg to protect it. GCN eggs can be distinguished from those of the other newts by their size, shape and colour. Submerged vegetation was examined for newt eggs and folded leaves gently opened to check for eggs. Once a GCN egg is identified, no further leaves need to be examined to minimise any further potential disturbance.

2.11 Appropriately licensed ecologists from FPCR Environment and Design Ltd completed all surveys and during suitable conditions.

Population Assessment

2.12 The population size class assessment as defined within the English Nature guidance (August 2001) is as follows:

Small – for maximum counts up to 10

Medium – for maximum counts between 11 – 100

Large – for maximum counts over 100

2.13 The above classification is a broad category used to indicate the comparative status and importance of a population, which has also informed consideration of the level of impact development may cause as well as having implications with regard to requirements for mitigation and compensation.

3.0 RESULTS AND EVALUATION

Field Survey

Pond Descriptions

3.1 There are a total of five ponds within the application site (Pond P1-4 and P6). A further pond is also present within the curtilage of South Marston Farm (Figure 1) although this pond was not accessed as part of the survey reported here.

Habitat Suitability Index

3.2 A Habitat Suitability Index (HSI) was calculated for each waterbody at the end of the season. Only a single pond (P2) had 'good' suitability for GCN with the remaining ponds scoring 'Average' (P1 and P3) or 'Below Average' (P4 and P6) suitability. Significant factors reducing the suitability of average or below average pond included the high level of shade caused by adjacent vegetation..

3.3 A summary of pond descriptions including HSI score and category is provided in Table 2 below.

Table 2: Habitat Suitability Index scores and pond suitability

Pond	Photo reference	Summary pond description	HSI Score	HSI Category
1		Heavily over-shaded pond with localised Callitriche sp, Veronica beccabunga, Equisetum palustre, Carex obtrubae	0.59	Average
2		Large pond with well-developed lush emergent vegetation with abundant Glyceria maxima, Rorippa nasturtium-aquatica, Carex hirta, Carex obtrubae, Juncus articulata, Callitriche stagnalis and water-crowfoot	0.77	Good

Pond	Photo reference	Summary pond description	HSI Score	HSI Category
3		Recently part-drained pond with exposed margins and part submerged tree stumps. Water crowfoot present **Fish present**	0.62	Average
4		Heavily over-shaded pond with sparse vegetation including Callitriche sp, Potentilla anserina, Juncus inflexus and ruderal vegetation.	0.51	Below average
5	No photo - Private land no access	Appears heavily over shaded by adjacent trees and shrubs	-	-
6		Over-shaded pond with engineered headwall at northern end. Abundant Lemna minor and occasional Iris psuedacorus, Rorippa nasturtium-aquatica and Carex obtrubae.	0.54	Below average
Off-site ponds in Nightingale wood		Dry ponds with Typha latifoliia, Eleocharis palustris and blanket weed. Some exposed silt present	-	-

Aquatic Surveys

3.4 Surveys completed within the Site confirmed the presence of great crested newt in two ponds (P1 and P6). As summary of results is provided in table 3 below:

Table 3: Aquatic survey result summary

Pond Number	Survey Date	Highest count									
		GCN		Smooth		Palmate		Toad		Frog	
		Num	Eggs	Num	Eggs	Num	Eggs	Num	eggs	Num	eggs
1	09/04/2015	-	-	4	-	-	-	-	-	-	-
	20/04/2015	1	-	-	-	-	-	-	-	-	-
	07/05/2015	2	Y	-	-	-	-	-	-	-	-
	14/05/2015	-	-	2	-	-	-	-	-	-	-
	08/06/2015	-	Y	1	-	-	-	-	-	-	-
	04/06/2015	-	Y	3	-	-	-	-	-	-	-
2	09/04/2015	-	-	3	-	-	-	-	-	-	-
	20/04/2015	-	-	7	Y	-	-	-	-	-	-
	07/05/2015	-	-	-	-	-	-	-	-	-	-
	14/05/2015	-	-	-	-	-	-	-	-	-	-
3	09/04/2015	-	-	-	-	-	-	-	-	11	-
	20/04/2015	-	-	-	-	-	-	-	-	-	-
	07/05/2015	-	-	-	-	-	-	-	-	-	-
	14/05/2015	-	-	-	-	-	-	-	-	-	-
4	09/04/2015	-	-	-	-	-	-	-	-	-	-
	20/04/2015	-	-	3	-	-	-	-	-	-	-
	07/05/2015	-	-	2	-	-	-	-	-	-	-
	14/05/2015	-	-	1	-	-	-	-	-	-	-
6	09/04/2015	3	Y	4	-	-	-	-	-	-	-
	20/04/2015	1	-	2	-	-	-	-	-	-	-
	07/05/2015	-	-	3	-	-	-	-	-	-	-
	14/05/2015	-	-	-	-	-	-	-	-	-	-
	08/06/2015	-	-	-	-	-	-	-	-	-	-
	04/06/2015	-	-	-	-	-	-	-	-	-	-

3.5 The survey data broadly supports that collected in previous years, which identified the presence of GCN in two ponds P1 and P4. GCN were recorded initially in P1 in 2002 (peak count 4) and, subsequently, in 2006 (peak count 1). The 2015 surveys yielded a peak count of 2 adult GCNs on the 7th May. GCN were first identified in P4, where a peak count of 9 individuals was recorded in 2006. No GCN were recorded from this pond during 2015.

3.6 In addition, GCN were also recorded in pond P6 in the north west of the site, where a maximum count of 3 adult GCN were observed on the 9 April 2015.

3.7 Offsite pond P5 was surveyed in 2002 and no evidence of GCN was observed

4.0 INTERPRETATION AND EVALUATION

4.1 The population size class assessment as defined within the English Nature guidance (August 2001) is as follows:

- Small – for maximum counts up to 10
- Medium – for maximum counts between 11 – 100
- Large – for maximum counts over 100

4.2 The above classification is a broad category used to indicate the comparative status and importance of a population.

4.3 Ponds P1 and P6 were observed to all support a small individuals populations of GCNs; as no more than three GCNs were found on any single survey occasion. The population does however appear to be low in both ponds.

4.4 The distance between ponds, in addition to the barrier formed by the South Marston Brook, is likely to indicate the two populations are distinct, with no or very little immigration and emigration between popiulations.

4.5 The current breeding ponds are generally suboptimal, both suffering from a lack of management that has lead to a high degree of over-shading and the development of deep silt/litter in the pond bases.

4.6 Adjacent commercial turf production and improved grassland areas, which forms the majority of habitat surrounding the ponds, are considered to be of low value depending on the management in operation. However, the more established hedgerows and coarse grassland and immediately adjacent habitats such as the plantation woodland around P1 are considered to be of moderate to high value especially given their potential value for shelter and foraging.

4.7 Previous surveys of the Site and wider area suggest that great crested newts (GCN) are fairly widespread but localised to the east and south of Swindon. These surveys identified the presence of a small population of GCN from a single pond associated with Lotmead Farm (SU 20135 85865); a small population in a pond to the south of Sainsbury's (SU 18799 86164) and a medium population from a cluster of ponds to the east of Dorcan (SU 19925 84237). The populations centred within the Site are both small and, in this context, are considered to be of no more that District significance.

5.0 PROPOSED HABITAT LOSS AND IMPACT ASSESSMENT

5.1 The presence of GCNs can potentially be a constraint on works because individual newts can use habitats up to 500m from a breeding pond[2]. English Nature (now Natural England) guidelines

[2] English Nature (2001) GCN Mitigation Guidelines, Peter borough (GCNMG)

state that suitable terrestrial habitats within 250 metres of a breeding pond are likely to be used most frequently by GCNs. More recent research commissioned by Natural England has shown that GCN densities are very low over 100m from the breeding pond, and that a majority occur within 50m of the pond[3]. This is also supported by previous research, which also suggests that where suitable habitat is present the majority of any GCN population will use terrestrial habitats within 50 metres of the breeding pond (see for example, Jehle (2003)[4].

5.2 Natural England, in order to aid the identification and assessment of impact, has identified three terrestrial zones: immediate (land within 50m of the breeding pond), intermediate (land within 50 and 250m of a breeding pond) and distant habitat (land over 250m from the breeding pond), which can be used to quantify impacts based on the value of habitat.

5.3 While the proposals have sought to retain ponds *in situ* as far as possible and as recommended by Natural England, the proposals would lead to the loss of areas of terrestrial habitat within the site. A summary of habitat loss is provided in table 4 below:

Table 4: Summary of terrestrial habitat impacts

Receptor	Habitat losses				Habitat to be retained and/or enhanced within site	
	Pond	Immediate terrestrial habitat (<50m from pond)	Intermediate terrestrial habitat (50-250m from pond)	Off-site (existing dwellings, roads, or habitat of low value (not secured for GCN))	Immediate terrestrial habitat (<50m from pond)	Intermediate terrestrial habitat (50-250m from pond)
Pond P1	No effect	0ha	13.8ha (of which 2.2ha is school)	2.5ha	0.8ha	5.1ha
Pond P6	Total loss	0.3ha	5.3ha	6.8ha	N/A	N/A

Pond P1 GCN Population

5.4 According to the Great Crested Newt Mitigation Guidelines the partial loss of intermediate habitat is of 'Low' significance with only the partial loss of immediate habitat being of 'Medium' impact. No high level impacts will occur.

5.5 The magnitude of terrestrial habitat loss in the absence of mitigation could lead to a potential decline in the conservation status of GCNs within the site and, potentially, their killing and injuring and disturbance of their places of rest or shelter during site clearance operations.

5.6 Additional impact may also occur as a result of the isolation/severance of ponds from the wider countryside leading to a reduction in the stability of any potential future wider metapopulation and

[3] Creswel and Whitworth, (2004). An assessment of efficiency of capture techniques and the value of different habitats for the GCN. Herpetological Journal vol. 10, p 143-5
[4] Jehle, R. (2000). The terrestrial summer habitat radio tracked GCNs (*Triturus cristatus* and marbled newts (*Triturus marmoratus*). The Herpetological Journal 10:137-143

populations within ponds. This could affect the ability of the population to respond to adverse environmental events.

Pond P6 GCN Population

5.7 The loss of the breeding pond and just under 50% of the immediate terrestrial habitat would lead to the loss of the population. A 'high' impact.

5.8 Consideration has been given to alternative scenarios such as the retention of this pond. However, due to its location close to existing built development and the requirement for new dwelling and roads any population would inevitably become isolated and therefore at greater risk from chance environmental events such as the pollution of the pond, which already appears to receive run-off from adjacent built development and consequently has a low water quality. Furthermore, it would become isolated from pond P4, which although not found to support GCN in the most recent survey has in recent years and a residual population may persist, albeit at low density. Isolation from this pond would further potentially reduce the stability of any metapopulation that may still occur. Translocation of GCN from P6 to an enhanced reserve that can be managed specifically for the benefit of GCN is therefore recommended.

6.0 MITIGATION AND ENHANCEMENT

EPS Licensing Requirement

6.1 Due to the loss of the breeding pond P6 and clearance of terrestrial habitats that could be used as resting/hibernation places by GCNs around both P1 and P6, it is considered that a Natural England European Protected Species (EPS) licence will need to be secured for the proposals to proceed. A detailed mitigation strategy (Method Statement) will also need to be produced in support of the license application. The mitigation approach will be to avoid killing or injuring GCNs and to minimise the impact on the GCN population in order to maintain their 'Favourable Conservation Status' in their natural range.

Mitigation Approach

6.2 The emphasis of mitigation will be to capture any newts present within the development zone, and relocate them to a receptor area, which has been specifically enhanced for that purpose.

6.3 It is the intention that the existing breeding ponds P1 is retained and enhanced. P4 would also be enhanced and additional ponds created in close proximity. Enhancements to ponds will include the thinning of over-shading vegetation and removal of silt and litter, which, in addition to the development of willow scrub within the ponds is currently threatening the permanence of the ponds.

6.4 Terrestrial habitat enhancement will include the planting of native scrub species and provision of hibernacula and refuge piles. Pond P1 also lies adjacent to a conifer plantation, which will be gradually thinned, with the objective of gradually removing the conifer stock and establishing a broad-leaved woodland in its place.

Receptor site

6.1 As part of the proposals large areas of tussocky grassland and newly-planted scrub are to be provided round both retained and created ponds. These areas will provide large areas of grassland with species including red clover *Trifolium pratense*, common bird's-foot trefoil *Lotus corniculatus*, ox-eye daisy *Leucanthemum vulgare*, common knapweed *Centaurea nigra* and common sorrel *Rumex acetosa* and new native tree and scrub planting. The existing hedgerows, where retained, are also to be enhanced and gapped up with native species including hazel *Corylus avellana*, holly *Ilex aquifolium*, guelder rose *Viburnum opulus*, crab apple *Malus sylvestris*, blackthorn *Prunus spinosa* and dog rose *Rosa arvensis)* along with the over seeding of margins adjacent to the hedgerows with native species grassland.

6.2 Similar species to those used to gap-up the hedgerows shall be used to provide scattered areas of scrub and woodland planting. Where grassland is currently present in retained areas the existing management regime will be amended, from relatively intensive agriculture to more sympathetic/relaxed management, to improve the botanical diversity and structure for amphibians.

6.3 Along with the above two hibernacula to be provided close to each of the breeding ponds. These hibernacula would follow the specifications as detailed within the Great Crested Newt Mitigation Guidelines (English Nature 2001) and would provide additional areas for shelter or refuge on site for GCN.

6.4 Existing ponds within the site will be restored to provide permanent open water. The results of the HSI would be used to inform suitable restoration and management practices for each pond, which could be expected to include

- the removal of choking vegetation where necessary to provide no more that 80% vegetation cover
- the removal of dense over-shading vegetation to no more that 50% cover
- The removal of silt from ponds suffering from early drying

Site Clearance

6.5 Trapping and translocation from habitats around P1 and P6 will only be carried out during the newt active season, ideally during the breeding migration period catching newts as they move to the breeding pond in spring.

6.6 Temporary amphibian fencing (TAF) with pitfall traps every 5 m will be installed within the site, with a particular focus of suitable habitat features present within 500m of the breeding pond. These areas will then be trapped out as specified in the Great Crested Newt Mitigation Guidelines for a small population (i.e. minimum of 30 nights trapping with a minimum trap density of 50 traps/ha). Upon completion of the trapping period appropriate potential refuge areas of the site shall be destructively searched under ecological supervision, prior to the commencement of works (any amphibians captured shall be translocated into the proposed receptor site).

Built Development

6.7 In order to ensure that any potential harm to GCN is reduced following the completion of the development, two alternative means of preventing impacts will be pursued if necessary. GCN will

either be prevented from entering the built development by the use of permanent amphibian exclusion fencing or, for instance, through modification of road or site surface water drainage infrastructure such as by the use of dropped kerbs and offset gullies to prevent GCN entrapment.

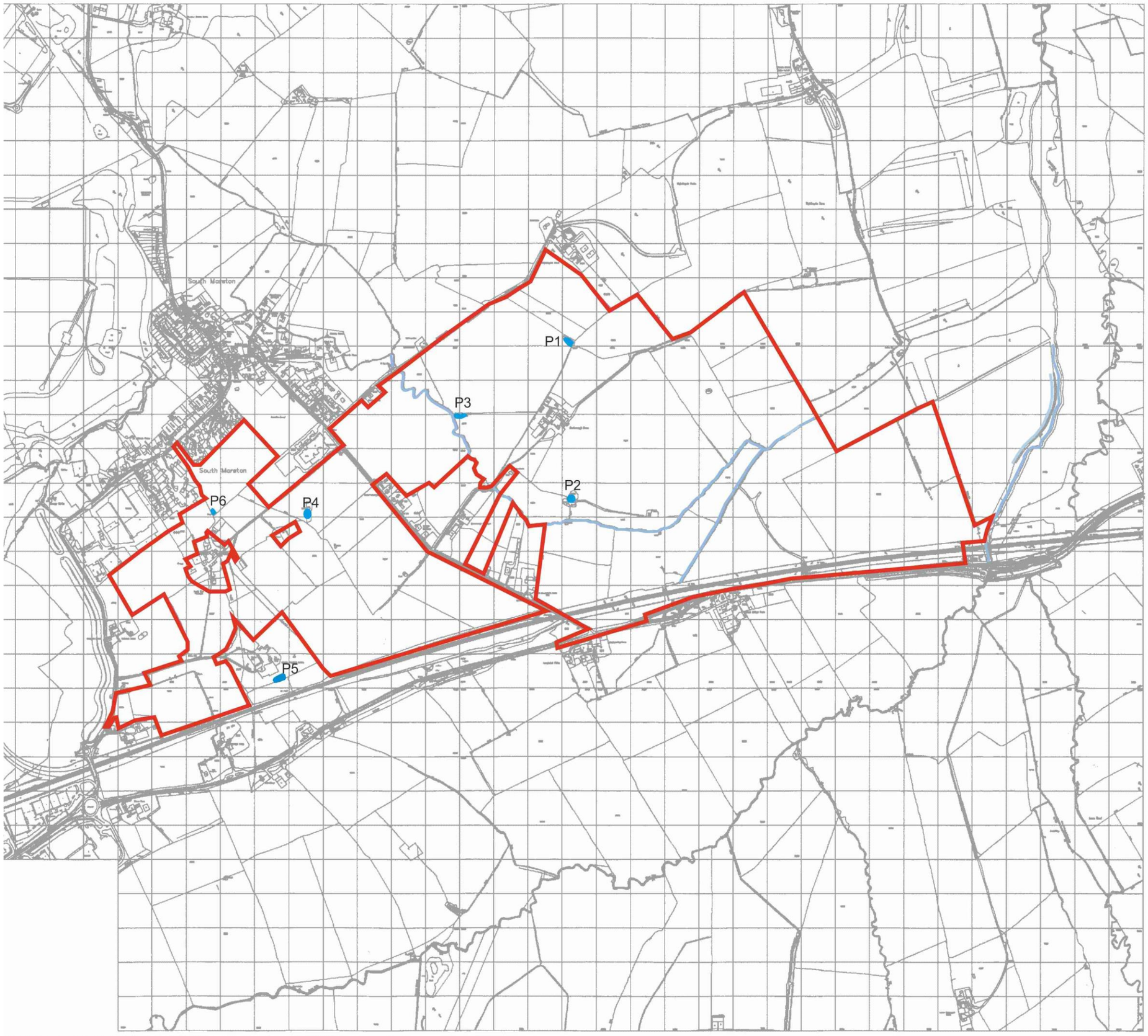
Monitoring

6.8 Post-development monitoring plays an important role in ensuring that mitigation proposals have worked. The GCNMG (and hence any Licence application) requires a two years monitoring to be undertaken for a 'Medium' impact on a small population to be completed post development to ensure that the favourable conservation status of the local population has been maintained.

7.0 CONCLUSIONS

7.1 This report presents compensation and mitigation measures that aim to maintain the local great crested newt population at a favourable conservation status. The loss of existing terrestrial habitat will be adequately compensated for by the creation optimal newt habitats around existing ponds and in the north and west of the site.

7.2 Great crested newts captured within the working areas prior to the commencement of construction operations will be relocated to this area, which will improve on the current situation and not only secure the future maintenance of the GCN population, which may be already suffering from decline as a result of the natural progression of some ponds from open water, through to, inevitably, dry land.



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Site Boundary

Pond (with reference)

Watercourse

Hallam Land Management, Taylor Wimpey & Hannick Homes

Eastern Villages, Swindon

POND PLAN

N

nts @ A3 PH / ph 14.10.2013

Figure 1 **1796-E—**

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